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PROJECT AND TEAM INFORMATION

Project Title

Dynamic Java based Quiz Application

Student/Team Information

Team Name: Team #	Bug-Hunters
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Bisht, Smriti – 240221598* *240221598@geu.ac.in*
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PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

The project is a java-based quiz Application designed to provide an interactive platform for users to attempt quizzes and evaluate their knowledge. The system supports user authentication, quiz attempts, real-time answer validation, and result generation. Administrator can manage question and quiz content efficiently.

The system is designed using Object-Oriented Programming principles to ensure modularity, scalability and maintainability. It follows a layered architecture separating the core logic, data management and user interface components. Key functionality includes question navigation, answer validation score calculation and performance tracking.

Updated Project Approach and Architecture (2 pts)

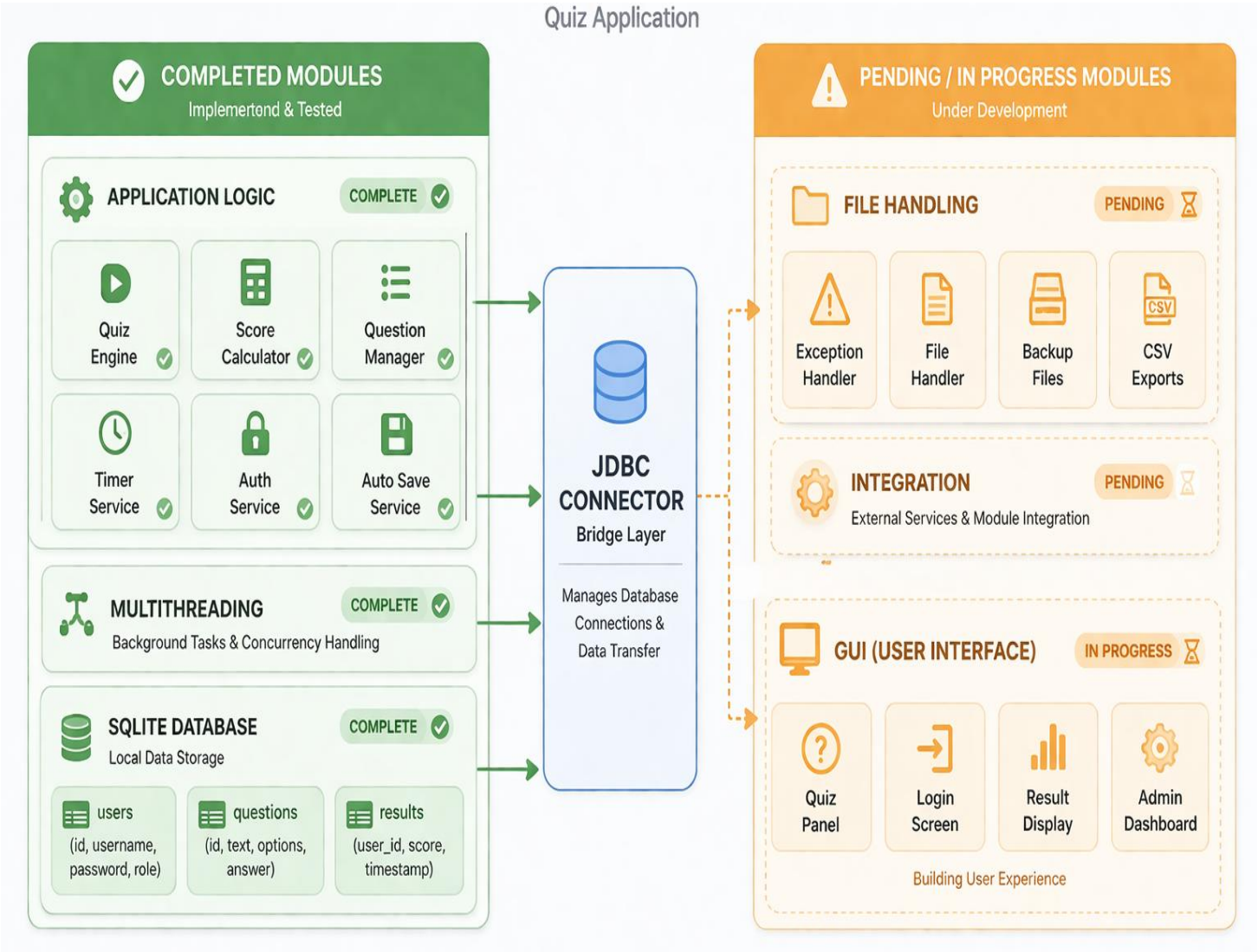
The project is being developed as a modular java desktop application using object-oriented Programming principles.

- **The architecture consists of:**
  - **Core logic Layer** – Handles quiz execution, scoring, and validation.
  - **Data Layer** - Responsible for storing questions (currently in memory, database planned).
  - **Presentation Layer (GUI)** – Under development using java FX.
- **Current implementation uses:**
  - Java collection for managing questions and answers
  - OOP concepts like encapsulation, inheritance, and abstraction
  - Modular class structure for scalability
- **Future integration will include:**
  - JDBC for database connectivity
  - File handling for import/export
  - GUI for user interaction

Tasks Completed (7 pts)

Task Completed	Team Member
Core Classes Design & Implementation	Smriti Bisht
Question Manager	Smriti Bisht
Quiz Logic	Smriti Bisht
User interface using JavaFX	Pranjal Chamoli
Database Integration	Tejas Kumar Rai
Multithreading	Pranjal Semwal

System Architecture



Challenges/Roadblocks (7 pts)

During development, several challenges were encounters:

1. **Design modular class structure**
   Ensuring proper separation of concerns between quiz logic and data handling required careful planning.
2. **Managing question flow and validation logic**
   Implementing correct answer checking and navigation between questions was initially complex.
3. **Integration planning**
   Since GUI, database and file handling are still pending, designing code that supports future integration without major refactoring is a challenge.

Solutions:

- Adopted layered architecture
- Used well-defined classes (Quiz, Question, Manager)
- Kept logic independent from UI and storage

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
File Handling (Import/Export)	Pranjal Semwal & Tejas Kumar
Module Integration	Smriti Bisht
Connecting to http clients for real time fetching	Smriti Bisht

Project Outcome/Deliverables (2 pts)

- Functional quiz engine with scoring system
- Question management system
- Modular java application structure
- GUI-based application (in progress)
- Optional database-supported persistent storage

Progress Overview (2 pts)

- Approximately 50-60% of the project is completed.
- **Completed:** core logic and backend modules
- **Pending:** File handling and integration
- **Optional:** Database module

The project is currently on track, with backend development ahead of schedule, while frontend development also being halfway done.

Codebase Information (2 pts)

- **Repository:** https://github.com/sumiiesque/quiz-application.git
- **Key Progress:**
  - Implemented Core classes
  - Developed quiz logic
  - Added question manager module
  - Added database integration
  - Added GUI modules

Testing and Validation Status (2 pts)

Test Type	Status (Pass/Fail)	Notes
Unit Testing (Quiz Logic)	Pass	Correct answer validation working
Question Handling	Pass	Questions stored and retrieved correctly
Integration testing	Pending	All UI modules along with the core modules work perfectly, database modules are yet to be tested.

Deliverables Progress (2 pts)

- **Core Logic Module → Completed**
- **Question Manager → Completed**
- **Quiz Engine → Completed**
- **GUI →In Progress**
- **File Handling →Pending**
- **Database → Completed**
- **Integration →Pending**